

Hints, Placebos and Shortcuts: Controlled Tests of Structural Primers for LLM Graph Reasoning

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Abstract

Prepending a *primer* of factual sentences about a graph’s structure — node degree, clustering coefficient, a random-walk structural encoding, or component membership — before its text encoding is a natural way to ease graph-reasoning prompts for a language model. Any apparent gain is illusory if a solver can recover the answer from the primer text without consulting the graph, and confounded if the primer merely lengthens the prompt. We evaluate four controls together: a graph-blind shortcut solver refit at the evaluation graph size, a length-matched content-free filler, a length-free content control, and a degree-preserving rewiring that varies triangle count while holding the prompt string length constant. On four arms (qwen3-1.7b, qwen3-4b, each with and without native thinking mode) at $n=40$, primer effects are organised chiefly by the model’s own no-primer accuracy: where the primer offers a substitute route to the answer, effect size falls with baseline ($r=-0.31$, 232 cross-fitted cells, $p=1.8\times 10^{-6}$); where it does not, there is no relation ($r=+0.02$, 762 cells). This reproduces on eleven held-out arms spanning 0.6–14B and a second model family ($r=-0.674$ over 105 route cells vs. $r=-0.186$ over 93 no-route cells) and, separately, under Game-of-Thrones node renaming ($r=0.933$ between the two naming schemes’ effect sizes for the route-offering primer). One non-answer-leaking effect — clustering on `node_degree` — beats the length-matched filler across five of seven density levels (+4.3pp pooled, $p=2.4\times 10^{-8}$) and, at high density where the larger model is off ceiling, replicates in qwen3-4b too (+11.3pp vs. none, +9.3pp vs. filler). A rewiring test that holds prompt length and degree sequence fixed while permuting triangle count does not show the benefit tracking the clustering values themselves. Two validity threats turn out to explain much of the rest of the sweep: `edge_existence` accu-

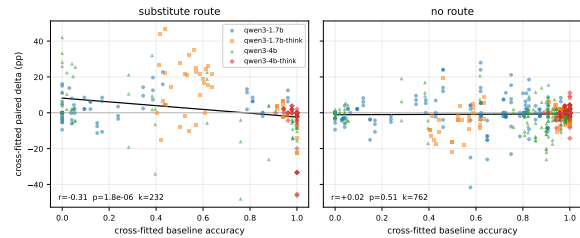


Figure 1: Cross-fitted paired delta vs. cross-fitted baseline accuracy (Section 5.4), for the 232 cells where a primer offers the model a substitute route to the answer (left) and the 762 that do not (right). Baseline and delta are estimated from disjoint halves of the paired graphs, so the negative slope on the left cannot be regression to the mean by construction.

racy moves mainly through a yes/no response bias (hit rate 99–100% throughout; false-alarm rate swinging 7.5–61.8% by condition), and `node_count`’s largest apparent gains fix an off-by-one artifact on a task whose gold answer is constant at this graph size. We report every result against a Benjamini–Hochberg correction and, for null cells, a minimum detectable effect, so an absence of significance is distinguished from a demonstrated absence of effect.

1 Introduction

Fatemi et al. (2023) show that a large language model’s accuracy on graph-reasoning questions depends substantially on how the graph is encoded as text, independent of reasoning ability, by varying the encoding function while holding the question and graph fixed. This work asks a different question about the same benchmark: does prepending a short *primer* of derived, node-level facts before the graph encoding help a model answer questions about that graph?

Two confounds make such an effect hard to read. If a primer helps because the model reads the answer out of the preamble rather than reasoning over the graph, that is answer leakage, not reasoning. If

it helps or hurts because it changes prompt length, that says nothing about content: Liu et al. (2024) and the needle-in-a-haystack protocol of Kamradt (2023) show that context length and position alone move retrieval accuracy. A third, quieter confound is measurement: an extraction bug that correlates with response style manifests as a condition effect with nothing in the resulting table that looks wrong (Section 3.4). A fourth is headroom: once a model is near ceiling on a task, no primer effect — helpful or harmful — has room to show up.

We build a control battery to separate these from a genuine structural effect: a graph-blind shortcut solver refit at the graph size actually evaluated (Section 3.2), a length-matched content-free *filler* condition, a length-free content control (*components*), and a degree-preserving rewiring that moves triangle count while holding the rendered prompt byte-identical in length (Section 3.3). Our contributions: (i) this control battery, applied together rather than one at a time; (ii) an empirical map of where primers move accuracy on $n=40$ Erdős–Rényi graphs, four model arms, six tasks; (iii) a demonstration that answer-stating primers act as a second route to the answer whose value is set by the model’s own baseline accuracy, with cross-fitted, cluster-bootstrapped evidence and two independent replications (held-out arms, node renaming); and (iv) one non-answer-leaking content effect, replicated within one model, across model scale in a baseline-matched regime, and constrained — not confirmed — by a rewiring test of its mechanism.

2 Related Work

We adopt the GraphQA task definitions of Fatemi et al. (2023) and fix graph size and edge density experimentally (Section 4) rather than sampling structure incidentally, so that difficulty is a manipulated variable. Wang et al. (2023) report LLMs are brittle to spurious correlations in graph and problem structure; Guo et al. (2023) find models struggle to exploit graph description languages directly; Jin et al. (2024) survey the wider LLM-graph literature. Dziri et al. (2023) argue transformers solve compositional tasks by linearised subgraph matching rather than systematic reasoning, which is the same question a primer’s apparent benefit raises: is a gain evidence of reasoning, or of a shorter pattern to match? Geirhos et al. (2020) frame this as *shortcut learning*; our shortcut

solver (Section 3.2) operationalises that concern for the primer channel specifically. The *rwse* condition adapts the random-walk structural encoding of Dwivedi et al. (2022), rendered as sentences rather than fed to an attention mechanism. Our degree-preserving rewiring (Section 3.3) follows the double-edge-swap configuration model of Foslom et al. (2018). Paired significance testing follows Dror et al. (2018). We distinguish the “thinking” arms evaluated here — Qwen3’s native thinking mode, toggled via the chat template — from chain-of-thought *prompting* (Wei et al., 2022); the two are not the same intervention, and we do not claim this paper tests the latter.

3 Method

3.1 Primers

A primer is a preamble of factual sentences about a graph’s local or global structure, prepended to the standard text encoding and the task question. A single renderer produces all seven conditions, so every condition shares identical sentence framing and formatting for the facts it states (the degenerate case, **filler**, states no fact and therefore uses its own frame, “Node N is simply present. . .”; this is a length control, not a content one, by design). **none** is the graph encoding alone. **components** gives one sentence with the graph’s connected-component count. **degree** states each node’s exact degree; **clustering** each node’s clustering coefficient; **rwse** each node’s random-walk return probabilities (Dwivedi et al., 2022). **all** combines degree, clustering and *rwse* (not *components*). Mean rendered length at $n=40$ ranges from 37 characters (*components*) to 4,691 (*all*); *filler* at 1,829 characters brackets *degree* (891) and *clustering* (1,629) from above but sits below *rwse* (2,949) and *all*, so it bounds rather than exactly matches the length term for those two conditions.

3.2 Shortcut solvers

A primer’s measured accuracy is evidence of graph reasoning only once it exceeds what a solver that never sees the graph can already recover from the primer text. Our shortcut solver combines sixteen exact rules, one heuristic, eight fitted rules (fit and scored on structurally disjoint graph sets), and an exact enumeration bound for small graphs; being structurally incapable of reading the graph object, its accuracy on a (task, condition) cell is a content-

Task (5–19 nodes)	none	comp.	clust.	rwse	degree	filler	all
node_count	.064	.076	1.00	1.00	1.00	1.00	1.00
edge_count	.018	.018	.148	.018	1.00	.018	1.00
node_degree	.082	.082	.082	.616	1.00	.082	1.00
connected_nodes	.082	.082	.082	.082	.208	.082	.352
edge_existence	.498	.498	.720	.642	.794	.498	.794
cycle_check	.832	1.00	.832	.832	.946	.832	.946
Task ($n=40$, mean over $p \in \{.10..85\}$)							
node_count	1.00	1.00	1.00	1.00	1.00	1.00	1.00
edge_count	.030	.030	.104	.030	1.00	.030	1.00
node_degree	.144	.144	.144	.210	1.00	.144	1.00
connected_nodes	.000	.000	.000	.001	.001	.000	.008
edge_existence	.735	.735	.727	.554	.746	.735	.741
cycle_check	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Table 1: Shortcut bars: accuracy of a solver that reads only the rendered primer, never the graph. Top: 500 graphs from the published generator (unchanged since these bars are the correct ones for the held-out arms). Bottom: refit at $n=40$, the corpus our main sweep evaluates, averaged over seven densities (per-density table in the Appendix). **Bold** marks cells a primer fully determines.

only lower bound, which we call the *shortcut bar* rather than a “ceiling” — it never overstates what a model could achieve, but does not itself upper-bound model accuracy.

The published GraphQA generator draws 5–19-node graphs; our evaluation corpus is $n=40$. A bar fit on the small-graph distribution does not transfer: refitting the same solver on $n=40$ Erdős-Rényi graphs at the seven densities used in our sweep (disjoint 300-graph fit/test sets per density) moves the `edge_existence/none` bar from 0.498 to a density-averaged 0.735, driven by class imbalance rather than anything a primer supplies, and drops the `degree-over-none` gap on that task from +0.296 to +0.011 — below the threshold at which we call a primer a substitute route. We therefore reclassify route status using the $n=40$ refit (Table 1), not the published split: only `node_degree` (under `rwse`, `degree`, `all`) and `edge_count` (under `clustering`, `degree`, `all`) remain routes once bars are measured on the right graph size; `edge_existence` and `connected_nodes` do not. `node_count` and `cycle_check` are excluded from the split entirely: at $n=40$ their gold answers are constant (40, and “yes” respectively, at every one of the seven densities we tested), so every condition’s bar sits near 1.0 under `none` already, and the published-split bars overstate what a primer adds here. The published-split bars (Table 1, top) remain the correct instrument for the held-out arms of Section 4, which answer that corpus.

3.3 Rewiring: isolating structure from length

The length-matched `filler` bounds the length confound but is not an exact match (Section 3.1), and no design that compares different graphs (or different generators) can hold a structural quantity fixed while claiming to vary only it: across five generator families at matched (n, m) , the majority-class baseline alone ranges 0.15 to 1.000. We add a within-instance design instead. A double-edge swap removes edges (a, b) , (c, d) and adds (a, d) , (c, b) (Fosdick et al., 2018); every endpoint keeps its degree exactly, so the degree sequence, n , m , and the rendered prompt’s character length are unchanged, while the triangle count — what clustering reports — moves. We generate low/base/high triangle-count graphs this way ($n=40$, $\bar{k}=12$, 200 graphs per level) and evaluate `node_degree` under `none`, `clustering` and `filler` on each, on qwen3-1.7b, its thinking twin, and qwen35-2b (Section 5.5).

3.4 Answer extraction and measurement validity

Turning free model text into a scored answer is a quiet source of measurement error, and it is not symmetric across conditions: thinking responses are longer and messier than zero-shot ones, so a naive extractor makes the thinking arm look worse than it is on exactly the comparison we want to draw. One failure is worth reporting in full because it produced a headline-sized effect that did not exist. Our boolean extractor originally took the last yes/no token in the response; on `cycle_check`, qwen3-4b frequently answers correctly and then recites the definition (“...a path with no repeated edges or nodes”), and the trailing “no” became the extracted answer. Correcting the extractor to ignore “no repeated” moved 225 of qwen3-4b’s scored boolean rows from wrong to right, all on `cycle_check`, none the other way; before the fix this read as a dramatic condition-dependent collapse, after it nothing on that task is significant. All numbers here use the corrected extractor. We also verified directly that Qwen3’s `<think>` tags never reach the extractor unstripped for the thinking arms (a spot check of 50 random rows per arm, plus a corpus-wide comparison of extraction with and without the pre-`</think>` text, finds no mismatch), and that scoring truncated (`hit_cap`) generations as wrong rather than dropping them changes the verdict on two

cells (rwse/cycle_check and clustering, all/cycle_check for qwen3-1.7b; full comparison in Appendix A). We drop truncated pairs throughout the main text and report the alternative there.

3.5 Significance testing

Every reported comparison is paired: a condition against none (or another condition) on the same graph and question, truncated pairs dropped from both sides. The primary test is the exact McNemar test (McNemar, 1947), Benjamini–Hochberg corrected (Benjamini and Hochberg, 1995) within each (arm, task) family of six conditions at $q=0.05$; where we report “beats both controls” we additionally require significance under a *global* BH correction pooling every test run for that claim, not only the family. For the baseline-dependence relation (Section 5.4) we use a cross-fitted design — baseline estimated from one random half of a cell’s paired graphs, delta from the disjoint other half, both directions pooled — with a cluster bootstrap that resamples whole (arm, task, density) blocks rather than individual cells, since conditions at the same block share one none sample. For null cells we report a minimum detectable effect (fast preset: 50 replicates, 200 permutations, 5 power-curve steps), so an absence of significance is distinguished from a demonstrated absence of effect rather than left ambiguous.

4 Experimental Setup

Tasks. Six GraphQA tasks (Fatemi et al., 2023): node count, edge count, node degree, connected nodes, edge existence, cycle check.

Models. We compare qwen3-1.7b and qwen3-4b (Qwen Team, 2025), each with native thinking mode on and off, for four arms. This pair was chosen for its paired thinking/non-thinking checkpoints at two sizes within one open-weight family and moderate compute cost; we make no claim about an architectural advantage for graph tasks specifically, and Section 5.4 extends the comparison to eleven further arms including a second family.

Sweep design. All four arms answer one shared prompt set: Erdős–Rényi graphs (Erdős and Rényi, 1959), $n=40$, four densities $p \in \{.10, .20, .35, .50\}$ (densfull40), 100 graphs per (density, task) cell across all seven conditions

and six tasks — 16,800 prompts per arm. Generation is greedy; the output budget is 2,048 tokens for the plain arms and 8,192 for the thinking arms (the two are not the same budget, and we say so whenever a comparison depends on it). A second corpus (densfull40hi) extends node_degree and edge_existence to $p \in \{.65, .75, .85\}$, 100 graphs/cell, all four arms. A dedicated node_degree sweep at 400 graphs/density level (all seven densities, qwen3-1.7b) backs the one content effect that survives every control (Section 5.5).

Held-out arms. Eleven further arms were not used to develop any claim here, though the corpus that eight of them answer — the published GraphQA split, 30 graphs/cell, five tasks, six conditions, plus a rerun of filler and edge_existence after a wording fix — pre-dates this project and was analysed earlier for other questions, so we call them earlier-collected rather than pre-registered. Eight arms (qwen3-8b, qwen3-14b, gemma-4-E4B-it, gemma-4-12B-it, each $\pm$ thinking) answer that split; three more (qwen3-0.6b $\pm$ thinking, qwen35-2b) answer a 100-graph {none, degree} probe over all six tasks. Together they extend the comparison to 0.6–14B and two families (Section 5.4).

Metrics. Exact match for every task, including connected_nodes: its set-F1 is 0.96–1.00 everywhere at this graph size, so it does not discriminate between conditions and exact match is what every table here uses.

Known validity caveats. node_count and cycle_check are shortcut-dominated at $n=40$ (Section 3.2); degree and all state the node_degree and edge_count answers verbatim. Results on these cells are reported but bound what can be concluded about graph reasoning rather than demonstrate it.

5 Results

All four arms completed the sweep (16,800 scored generations each). Unless stated otherwise, Δ is a paired difference against none over instances where neither arm was truncated.

Task	1.7B	1.7B-th.	4B	4B-th.
edge_count	22.4%	81.0%	0.6%	62.2%
cycle_check	6.1%	6.5%	1.0%	1.9%
node_degree	0.0%	1.1%	0.0%	1.6%
connected_nodes	0.0%	0.8%	0.0%	3.5%
edge_existence	0.0%	0.2%	0.1%	0.8%
node_count	0.0%	2.5%	0.0%	0.1%

Table 2: Truncation rate (share of `densfull40` generations reaching the output budget: 2,048 tokens plain, 8,192 thinking), by task and arm, pooled over condition and density.

5.1 Validity first: truncation and constant answers

Truncation (Table 2) is neither uniform across tasks nor arms: on `edge_count` the thinking arms truncate on 62–81% of instances, so surviving pairs there are a biased subset and we report but do not interpret that cell under thinking. Every other task truncates under 7%. Separately, `node_count` (gold = 40, constant) and `cycle_check` (gold = “yes” at every density we tested) are shortcut-dominated (Section 3.2); apparent gains there are reported in Section 5.2 but read as an artifact (Section 5.3), not as graph reasoning.

5.2 The full density sweep

Table 3 (thinking arms in the Appendix, Table 4) pools over the four densities. Bold marks significance under McNemar, BH-corrected within each (arm, task) family. Several cells are significant in one arm without replicating in the other at the same sign and magnitude; Section 5.4 explains why that is not evidence against a structural effect, since the two model sizes are rarely at matched baseline on the same cell.

5.3 Length is an active treatment

`filler` is not inert. It significantly harms `qwen3-1.7b` on `edge_existence` (−15.8) and `connected_nodes` (−6.2), and `qwen3-4b` on both (−7.0, −12.5). Over the full seven-density range (`qwen3-1.7b-think`, `node_degree`) it costs −6.3 points pooled ($p < 10^{-4}$): negligible at $p \leq .35$, then −9.5, −9.3, −14.1, −11.7 at the four densest levels ($r = +0.953$, $p = 0.00088$ against baseline). Length’s cost grows with task difficulty, the same direction an answer-stating primer’s benefit grows in (Section 5.4); pooling all conditions into one correlation against `none` would understate both.

The largest `edge_existence` swings in Table 3 are not graph reading. Table 5 decomposes accuracy into hit rate and false-alarm rate: hit rate sits at 99–100% under every condition for both arms, so the model essentially never misses a real edge. What moves is the false-alarm rate — how often it says “yes” to a non-edge — and it moves by condition, not by anything resembling graph content: `filler` raises it (40.3→61.8%, 12.6→22.5%) while `clustering` lowers it. We read `edge_existence` condition effects as shifting a yes/no response bias, not as evidence of reading the graph more or less accurately.

`node_count`’s largest apparent gains are the same kind of artifact. Gold is the constant 40 at this graph size; the dominant `none` failure is the answer “39”, an off-by-one from the encoder’s “nodes 0, ..., 39” preamble. `qwen3-1.7b`’s “39” rate is 68.5% under `none`, 61.8% under `degree`, 28.8% under `filler`, and 1.3% under `clustering` — almost any interposed text reduces it, `clustering` nearly eliminates it, and none of this is a structural primer effect on a task with one possible answer.

5.4 Answer-stating primers: route substitution

Table 6 splits the 354 non-degenerate cells by whether the shortcut solver, refit at $n=40$, recovers more from the condition than from `none`. Where it does, effect falls with baseline at $r = -0.417$ ($p = 1.1 \times 10^{-4}$); where it does not, there is no relation ($r = -0.050$, $p = 0.41$). Because this split’s two groups are not baseline-matched, we treat Figure 1 — baseline and delta estimated from disjoint graph halves, so a low-baseline pair cannot mechanically inflate its own delta — as the stronger evidence: $r = -0.31$ over 232 route cells ($p = 1.8 \times 10^{-6}$, 95% cluster-bootstrap CI [−0.42, −0.17]) against $r = +0.02$ over 762 no-route cells ($p = 0.51$). A mixed-effects model with a random intercept per arm gives the same sign and a comparable magnitude on the route cells (−12.0, $p = 1.2 \times 10^{-5}$; only four arms, so we read this as a consistency check, not an independently powered test).

The density continuum makes the same point with difficulty manipulated rather than observed. Holding the arm (`qwen3-1.7b-think`) and task (`node_degree`) fixed and varying only density from $p = .10$ to $.85$ drives `none` accuracy from 0.95 to 0.44; `degree`’s benefit rises from +3.1 to +34.2 points ($r = -0.975$, $p = 1.9 \times 10^{-4}$), while

Arm	Task	none	comp.	clust.	rwse	degree	filler	all
qwen3-1.7b	connected_nodes	72.5	+0.8	+1.2	+1.0	+3.5	-6.2	+2.0
	cycle_check	99.7	-0.6	+0.3	-7.5	-12.6	+0.3	-1.1
	edge_count	1.8	-1.4	-0.4	-0.8	+4.0	-0.4	+2.4
	edge_existence	70.5	-8.2	+3.8	+5.3	+4.0	-15.8	+11.5
	node_count	31.5	+21.2	+67.2	+2.0	+6.8	+39.8	+47.0
	node_degree	60.5	+1.8	+3.0	+4.0	+7.5	-0.5	+10.2
qwen3-4b	connected_nodes	86.0	+9.2	+1.0	+0.0	-3.0	-12.5	-2.8
	cycle_check	99.7	-1.3	+0.3	+0.3	-1.5	-0.3	-0.5
	edge_count	2.0	-1.3	-1.3	-1.5	+27.7	+1.2	+3.8
	edge_existence	90.5	-2.8	+4.0	-10.0	-4.3	-7.0	-9.8
	node_count	90.2	+9.0	+8.8	+9.2	+8.0	+9.2	+9.8
	node_degree	99.2	+0.2	-0.5	-2.2	-6.5	-0.5	-8.0

Table 3: Main sweep, plain arms (thinking arms in Table 4), $n=40$, pooled over densities $p \in \{0.10, 0.20, 0.35, 0.50\}$, 100 graphs per (task, primer, density) cell. The none column is absolute accuracy (%); all other columns are paired differences against none in percentage points. Truncated generations excluded pairwise, $n \in [211, 400]$ paired instances/cell. **Bold**: significant, exact McNemar, Benjamini–Hochberg within each (arm, task) family, $q = 0.05$.

Arm	Task	none	comp.	clust.	rwse	degree	filler	all
qwen3-1.7b-think	connected_nodes	79.7	+2.8	+1.5	-0.3	+7.4	+2.0	+6.1
	cycle_check	100.0	+0.0	+0.0	+0.0	-0.7	+0.0	+0.0
	edge_count	86.3	+7.4	-8.7	-4.0	-61.5	+3.4	-8.3
	edge_existence	99.2	-0.8	-1.0	-1.3	-2.3	-0.8	-2.0
	node_count	98.6	+0.9	+1.4	+1.4	+0.8	+1.4	+1.4
	node_degree	76.8	-1.0	+1.0	+2.5	+11.7	-3.5	+11.6
qwen3-4b-think	connected_nodes	98.7	+0.5	+0.0	+0.0	+0.0	+0.8	-0.5
	cycle_check	100.0	+0.0	+0.0	+0.0	-0.8	+0.0	+0.0
	edge_count	97.6	+1.7	+1.7	+0.0	-23.5	-2.0	-41.5
	edge_existence	100.0	+0.0	+0.0	+0.0	+0.0	+0.0	+0.0
	node_count	99.5	+0.0	+0.5	+0.5	+0.5	+0.5	+0.5
	node_degree	99.7	-0.3	-0.3	-2.1	+0.0	+0.0	-0.8

Table 4: Main sweep, thinking arms; conventions as in Table 3. Both arms are at or near ceiling on most tasks, leaving no headroom for a primer effect. edge_count truncates on 62–81% of rows under thinking (Table 2) and is reported but not interpreted. $n \in [13, 400]$ paired instances/cell. **Bold**: significant, exact McNemar, Benjamini–Hochberg within each (arm, task) family, $q = 0.05$.

Arm	Cond.	hit rate	false alarm
qwen3-1.7b	none	100.0%	40.3%
	filler	100.0%	61.8%
	clustering	100.0%	35.2%
qwen3-4b	none	99.1%	12.6%
	filler	100.0%	22.5%
	clustering	100.0%	7.5%

Table 5: edge_existence, $p=0.50$: hit rate (correct on gold “yes”) vs. false-alarm rate (wrong “yes” on gold “no”). Accuracy moves almost entirely through the false-alarm rate.

filler’s cost grows from negligible to -14.1 ($r=+0.953$, $p=8.8 \times 10^{-4}$). The two effects run in opposite directions and grow together as the task hardens.

Held-out arms and a second family. On the eleven held-out arms (including, unlike an

earlier pass, the .rerun rows for filler and edge_existence), the relation reappears where a route exists: $r=-0.674$ over 105 cells ($p \approx 0$), against $r=-0.186$ over 93 no-route cells ($p=0.075$). Regressing effect on baseline, log parameter count and thinking mode together gives a baseline coefficient of -30.2 ($p=3.5 \times 10^{-15}$) and a parameter coefficient of $+0.40$ ($p=0.54$): once baseline is in the model, scale adds nothing measurable. Replacing each cell’s own baseline with the other arms’ accuracy on the same cell — a negative control sharing no observation with the effect — leaves no relation ($r=+0.113$, $p=0.135$), so what the effect tracks is a model’s own competence on an item, not a difficulty intrinsic to the item.

Renaming nodes. If the route effect were token copying rather than content substitution, renaming nodes from integers to Game-of-Thrones characters throughout the prompt should disrupt it. It does not:

Accuracy under	substitute route		no route	
none	cells	Δ	cells	Δ
0.00–0.25	24	+6.3	24	−0.7
0.25–0.50	6	+8.5	6	+2.0
0.50–0.75	6	+15.5	24	+0.4
0.75–0.95	11	+4.2	33	−3.9
0.95–1.00	34	−9.0	186	−0.4
$r(\Delta, \text{baseline})$	−0.417		−0.050	
	$k=81, p=1.1\times 10^{-4}$		$k=273, p=0.41$	

Table 6: Mean primer effect in percentage points, binned by `none` accuracy on the same cell, using the $n=40$ -refit route split (Table 1, bottom). Route baseline 0.614 ± 0.428 ; no-route 0.845 ± 0.291 — the groups are no longer as closely baseline-matched as an earlier, published-split classification made them look, which Figure 1’s cross-fitted design is the more decisive evidence for.

on the published split, the route-offering `degree` condition’s effect size correlates $r=0.933$ between the integer- and GoT-named sweeps (39 cells), while non-route conditions correlate weakly or not at all (`clustering` $r=0.079$, `components` $r=0.345$, `rwse` $r=-0.381$). Separately, renaming nodes with no primer at all changes accuracy by a precise, non-significant -0.19 points pooled across eight arms (zero of eight significant after BH), so the route-transfer result is not naming changing accuracy on its own, but the same substitution effect surviving a token-level change to how the answer is expressed.

What copying looks like. Where a primer states the answer and the model still errs, 41.4% of `qwen3-4b`’s 29 `degree/node_degree` errors state an adjacent node’s degree ($k\pm 1$) rather than the queried node’s, against 5.7% under `all`, which also states the answer but states thirty-nine other facts alongside it. This is consistent with position-sensitive copying under `degree` specifically, not a general degradation from stating the answer.

Practical reading. A primer effect reported against a no-primer baseline, without reporting that baseline, is uninterpretable: within `qwen3-4b` alone, `degree` is worth $+27.7$ on `edge_count` (baseline 0.02) and -6.5 on `node_degree` (baseline 0.99) — a cross-task, not a same-task, contrast, since the `degree` primer states `node_degree`’s answer verbatim while only implying `edge_count`’s through a sum, but the direction is the same one the baseline relation predicts.

Comparison	Δ	n	density	p
vs. none, main sweep	+3.0	400	.10–.50	n.s.
vs. none, 7 densities	+1.9	2800	.10–.85	0.022
vs. filler, 7 densities	+4.3	2800	.10–.85	2.4×10^{-8}
vs. none, replication	+4.2	1600	.10–.50	0.00055

Table 7: `qwen3-1.7b`, `node_degree`, `clustering`. The dedicated 400-graph/density sweep’s own filler arm covers all seven densities, unlike `densfull140`’s.

5.5 Non-answer content: clustering on `node_degree`

Table 7 is the one cell we find robust to every control we have. Against `none` alone, pooling all seven densities washes the effect down to $+1.9$ ($p=0.022$): it is positive and significant at $p=.35$ ($+7.5$), null at the two lowest and two highest densities where `none` itself is near floor, and briefly negative on one density. Against `filler` the picture is cleaner and holds where the `none` comparison does not: $+4.3$ pooled ($p=2.4\times 10^{-8}$), significant at five of seven densities, and negative only at $p=.10$ (-2.8 , $p=0.043$, where the model is already near ceiling). Churn is real but the effect survives it: 53 instances helped, 41 hurt on the main sweep alone. `qwen3-4b` cannot test this cell in the main densities (baseline 0.99), but at high density (`densfull140hi`, $p\in\{.65, .75, .85\}$, where its own baseline drops to headroom) `clustering` beats `none` and `filler` and `components`, all globally BH-significant: $+11.3$ vs. `none` ($p<10^{-3}$), $+9.3$ vs. `filler` ($p=6\times 10^{-4}$), on `node_degree`, and $+5.3/+11.7$ on `edge_existence`. This is the cross-scale replication a blanket “no primer effect replicates across model scale” claim would miss: the effect does replicate once the second model is measured off ceiling, at the baseline where it has room to show up. `qwen3-1.7b-think` does not show it (Table 4, Appendix), and `qwen3-4b-think` is saturated on this task throughout.

The mechanism we can rule in is limited. Figure 2 holds every arm’s degree sequence, n , m and rendered prompt length fixed within triangle level and varies only triangle count (Section 3.3). `clustering`’s curve moves roughly in parallel with `none`’s own level-to-level shape rather than tracking the actual clustering values, which fall monotonically from low to high by construction while the benefit

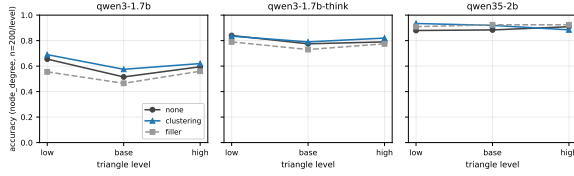


Figure 2: node_degree accuracy by triangle level (degree sequence, n , m and prompt length held fixed within each arm) and condition, $n=200/\text{level}$, $n=40\text{-k12}$ rung.

does not: for qwen3-1.7b, clustering—none is +3.5/+6.0/+2.5 at low/base/high. For qwen35-2b, filler matches or beats clustering at every level. We read this as evidence the clustering benefit is not simply a function of the clustering values stated, and leave the mechanism as future work (Section 8) rather than infer one.

5.6 Thinking mode

Thinking moves most (task, arm) cells to ceiling, leaving no headroom for a primer effect, and where it does not saturate a task it can censor it instead (Table 2). It also changes how long the model generates before answering, which is itself condition-dependent: qwen3-1.7b-think’s degree raises the non-termination rate on node_degree by 3.4pp over none ($p<0.001$) while filler lowers it; outside edge_count, degree produces 137 non-terminating generations of which 7 also fail to terminate under none — 130 new failures attributable to the primer. Separately, qwen3-4b-think answers correctly on all 29 of the instances degree breaks for qwen3-4b, under the identical primer and weights: whatever the primer disrupts is recoverable within the same network once a reasoning trace runs, which we report as an observation about the process, not as evidence for a specific mechanism. We do not report a single pooled “thinking beats every primer effect by an order of magnitude” figure: the comparisons available differ in token budget (2,048 vs. 8,192) and task mix, so a fair same-budget comparison is not one we have.

5.7 Earlier-collected models

The eleven held-out arms are mostly near ceiling under none on the published split (mean 0.924; gemma4-12b 0.989, qwen3-8b-think 1.000), so headroom, not primer lift, is the question they answer best; only the smallest arms

(qwen3-0.6b 0.705, qwen35-2b 0.876) have meaningful room. A confirmatory 500-graph replication (qwen3-8b, degree, pre-specified condition) gives +6.67pp pooled ($p=1\times 10^{-4}$), one instance of route substitution at 8B. Renaming nodes to Game-of-Thrones characters with no primer moves accuracy by -0.19pp pooled, zero of eight arms significant after BH (Section 5.4), a precise null over 1,200+ paired rows per arm.

5.8 Power

For every null cell in the main sweep we compute a minimum detectable effect (Section 3.5) rather than report only a p -value. Coverage is uneven: on qwen3-1.7b and qwen3-4b 10 of 24/20 null cells cannot detect even a 1.0-point effect at our power target; on qwen3-4b-think, near ceiling on most tasks, all 33 null cells are underpowered at that threshold, which is the ceiling speaking, not evidence the primer has no effect there. Full table in the Appendix.

6 Discussion

Three regularities account for most of what we measure, and only one of them is a content effect. First, a primer that states the requested quantity acts as a second route to the answer, and whether that route helps is set by whether it is more reliable than the model’s own — a relation that holds across four cross-checks (cross-fitted cells, eleven held-out arms, a negative-control difficulty estimate, and node renaming) and needs none of them individually to be conclusive. Second, added text is an active treatment regardless of content: the length-matched filler harms both model sizes on some tasks, at magnitudes exceeding most content effects measured here, and its cost grows with task difficulty in the same way an answer-stating primer’s benefit does. An experimental design that compares a primer only against none estimates content and length confounded. Third, headroom is scarce and unevenly distributed: thinking saturates most cells, class imbalance inflates edge_existence’s blind-solver bar at this graph size, and a fixed gold answer removes node_count and cycle_check from consideration entirely.

Against that background, one content effect survives: clustering on node_degree beats a length control across most of the density range in one model, and replicates in a second, larger model

once that model is evaluated off its own ceiling. We read this as a genuine finding about where headroom and this content overlap, not as evidence for a mechanism: the rewiring test that could show the benefit tracking triangle count directly does not show that, and we report the negative result alongside the positive one rather than the positive one alone.

We emphasise throughout that these are behavioural claims. We measured no attention weights, activations or internal states, and did not intervene on the model’s computation.

7 Limitations

The main sweep covers two model sizes from one family; the held-out arms extend the range but on a smaller, earlier-collected corpus with fewer conditions, and were not pre-registered against this project’s hypotheses. The route/no-route split depends on a threshold (Section 3.2) and on the $n=40$ shortcut refit’s own fit/test split; we report the reclassification it produced relative to the published-split bars as a sensitivity check rather than treating either bar set as definitive. Graphs are drawn from one generator family (Erdős–Rényi); the rewiring test varies triangle count within that family but does not test transfer to another. The length control (`filler`) brackets rather than exactly matches length for two of five content conditions. The crossover baseline at which an answer-stating primer stops helping does not transfer between corpora (near 0.7 at $n=40$ on the route-only split, near 0.99 on the published split), so it is a property of a corpus, not a universal constant. Finally, several results in Section 5.5 and 5.7 rest on $n \leq 300$ per cell; we report a minimum detectable effect for every null result but that does not substitute for a larger sample where one would change the conclusion.

8 Future Work

Two prompt-only experiments would discriminate directly between the accounts in Section 6 without new machinery. A *single-fact* primer stating only the queried node’s degree keeps the answer available while removing the other 39 sentences, separating interference from route substitution. A *corrupted* primer stating a wrong degree measures reliance directly: a model that copies collapses, a model that computes is unaffected. Extending the clustering-on-node_degree sweep and the rewiring

test to `qwen3-4b` at high density, and down and up the Qwen3 parameter ladder, would turn the two-point cross-scale replication in Section 5.5 into a curve. The rewiring result itself invites a direct test of what *does* predict the clustering benefit, since triangle count alone does not appear to.

Ethics Statement

This work evaluates open-weight language models on synthetically generated Erdős–Rényi graphs. It involves no human subjects, no personal or identifying data, and no annotation labour. We see no direct risk of harm from the findings; the principal societal cost is the compute expended.

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A Supplementary tables

Paired effect sizes for every (arm, task, primer) cell: Δ in percentage points against none, a 95% paired-bootstrap interval, the permutation p -value, and Δ under the alternative convention that scores truncated generations incorrect rather than dropping the pair (Section 3.4). n is the number of pairs in which neither arm truncated.

Task	Primer	n	Δ	95% CI	perm p	zeroed
connected_nodes	components	400	+0.8	[−2.8, +4.5]	0.7861	+0.8
	clustering	400	+1.2	[−3.0, +5.8]	0.6458	+1.2
	rwse	400	+1.0	[−3.5, +5.2]	0.7304	+1.0
	degree	400	+3.5	[−0.5, +7.5]	0.1239	+3.5
	filler	400	−6.2	[−10.2, −2.2]	0.0040	−6.2
	all	399	+2.0	[−2.8, +6.8]	0.4644	+2.0
cycle_check	components	343	−0.6	[−1.5, +0.0]	0.5033	+2.0
	clustering	362	+0.3	[+0.0, +0.8]	1.0000	+8.5
	rwse	358	−7.5	[−10.3, −4.7]	0.0001	−0.2
	degree	301	−12.6	[−16.6, −9.0]	0.0001	−17.5
	filler	341	+0.3	[+0.0, +0.9]	1.0000	+3.0
	all	362	−1.1	[−2.5, +0.0]	0.2210	+7.0
edge_count	components	211	−1.4	[−3.8, +0.5]	0.3783	−1.0
	clustering	238	−0.4	[−2.5, +1.7]	1.0000	−0.5
	rwse	239	−0.8	[−2.9, +1.3]	0.6837	−0.8
	degree	253	+4.0	[+0.8, +7.1]	0.0179	+3.0
	filler	224	−0.4	[−2.2, +1.3]	1.0000	−0.5
	all	211	+2.4	[−0.5, +5.7]	0.2269	+0.8
edge_existence	components	400	−8.2	[−11.8, −4.8]	0.0001	−8.2
	clustering	400	+3.8	[−0.5, +8.0]	0.1199	+3.8
	rwse	399	+5.3	[+0.8, +9.8]	0.0321	+5.0
	degree	400	+4.0	[−0.5, +8.2]	0.0957	+4.0
	filler	400	−15.8	[−20.0, −11.8]	0.0001	−15.8
	all	400	+11.5	[+6.8, +16.2]	0.0001	+11.5
node_count	components	400	+21.2	[+15.8, +26.8]	0.0001	+21.2
	clustering	400	+67.2	[+62.7, +71.8]	0.0001	+67.2
	rwse	400	+2.0	[−4.2, +8.5]	0.5871	+2.0
	degree	400	+6.8	[+3.0, +10.5]	0.0008	+6.8
	filler	400	+39.8	[+35.0, +44.8]	0.0001	+39.8
	all	400	+47.0	[+42.0, +52.0]	0.0001	+47.0
node_degree	components	400	+1.8	[−2.0, +5.8]	0.4599	+1.8
	clustering	400	+3.0	[−1.8, +7.8]	0.2577	+3.0
	rwse	399	+4.0	[−0.5, +8.5]	0.1104	+4.0
	degree	400	+7.5	[+3.0, +12.0]	0.0018	+7.5
	filler	400	−0.5	[−4.5, +3.5]	0.9028	−0.5
	all	400	+10.2	[+5.2, +15.2]	0.0003	+10.2

Table 8: qwen3-1.7b: all cells.

Task	Primer	n	Δ	95% CI	perm p	zeroed
connected_nodes	components	392	+2.8	[−0.5, +6.1]	0.1268	+2.5
	clustering	394	+1.5	[−2.5, +5.6]	0.5384	+2.0
	rwse	393	−0.3	[−4.1, +3.6]	1.0000	+0.2
	degree	390	+7.4	[+3.6, +11.3]	0.0001	+6.2
	filler	394	+2.0	[−1.5, +5.6]	0.3275	+2.5
	all	392	+6.1	[+2.6, +9.9]	0.0027	+6.2
cycle_check	components	346	+0.0	[+0.0, +0.0]	1.0000	−7.5
	clustering	376	+0.0	[+0.0, +0.0]	1.0000	−0.5
	rwse	386	+0.0	[+0.0, +0.0]	1.0000	+2.2
	degree	295	−0.7	[−1.7, +0.0]	0.5036	−21.8
	filler	388	+0.0	[+0.0, +0.0]	1.0000	+3.0
	all	373	+0.0	[+0.0, +0.0]	1.0000	−0.8
edge_count	components	27	+7.4	[+0.0, +18.5]	0.5037	−0.8
	clustering	23	−8.7	[−26.1, +8.7]	0.6251	−1.2
	rwse	25	−4.0	[−12.0, +0.0]	1.0000	−3.2
	degree	13	−61.5	[−84.6, −38.5]	0.0102	−3.2
	filler	29	+3.4	[−6.9, +13.8]	1.0000	−1.5
	all	24	−8.3	[−29.2, +12.5]	0.6794	+5.0
edge_existence	components	398	−0.8	[−2.3, +0.8]	0.5090	−0.8
	clustering	398	−1.0	[−2.5, +0.3]	0.2911	−1.0
	rwse	398	−1.3	[−3.0, +0.3]	0.2274	−1.2
	degree	397	−2.3	[−4.3, −0.5]	0.0249	−2.5
	filler	399	−0.8	[−2.3, +0.8]	0.5051	−0.5
	all	399	−2.0	[−3.8, −0.5]	0.0374	−1.8
node_count	components	342	+0.9	[−0.3, +2.3]	0.3758	+1.5
	clustering	369	+1.4	[+0.3, +2.7]	0.0597	+9.0
	rwse	369	+1.4	[+0.3, +2.7]	0.0597	+9.0
	degree	360	+0.8	[−0.6, +2.2]	0.4512	+5.5
	filler	369	+1.4	[+0.3, +2.7]	0.0597	+9.0
	all	369	+1.4	[+0.3, +2.7]	0.0597	+9.0
node_degree	components	395	−1.0	[−4.3, +2.3]	0.6543	−1.0
	clustering	396	+1.0	[−2.8, +4.8]	0.7010	+1.5
	rwse	394	+2.5	[−1.5, +6.9]	0.2898	+2.5
	degree	376	+11.7	[+7.4, +16.0]	0.0001	+8.8
	filler	396	−3.5	[−7.3, +0.3]	0.0954	−3.0
	all	396	+11.6	[+7.3, +15.9]	0.0001	+12.2

Table 9: qwen3-1.7b-think: all cells.

Task	Primer	n	Δ	95% CI	perm p	zeroed
connected_nodes	components	400	+9.2	[+6.0, +12.8]	0.0001	+9.2
	clustering	400	+1.0	[-3.0, +4.8]	0.6893	+1.0
	rwse	400	+0.0	[-4.0, +4.0]	1.0000	+0.0
	degree	400	-3.0	[-7.2, +1.2]	0.1882	-3.0
	filler	400	-12.5	[-17.0, -8.0]	0.0001	-12.5
	all	400	-2.8	[-7.0, +1.5]	0.2511	-2.8
cycle_check	components	391	-1.3	[-2.6, +0.0]	0.1318	-2.0
	clustering	395	+0.3	[+0.0, +0.8]	1.0000	+0.5
	rwse	392	+0.3	[+0.0, +0.8]	1.0000	-0.2
	degree	391	-1.5	[-2.8, -0.5]	0.0312	-2.2
	filler	392	-0.3	[-0.8, +0.0]	1.0000	-0.5
	all	395	-0.5	[-1.5, +0.5]	0.6269	-0.2
edge_count	components	397	-1.3	[-2.8, +0.3]	0.1756	-1.2
	clustering	399	-1.3	[-3.0, +0.3]	0.2298	-1.2
	rwse	398	-1.5	[-3.0, -0.3]	0.0668	-1.5
	degree	393	+27.7	[+23.2, +32.6]	0.0001	+27.3
	filler	400	+1.2	[-0.5, +3.2]	0.3012	+1.2
	all	397	+3.8	[+1.3, +6.3]	0.0049	+3.8
edge_existence	components	400	-2.8	[-6.0, +0.5]	0.1254	-2.8
	clustering	400	+4.0	[+1.0, +7.0]	0.0168	+4.0
	rwse	400	-10.0	[-14.0, -6.2]	0.0001	-10.0
	degree	398	-4.3	[-7.8, -1.0]	0.0200	-4.8
	filler	400	-7.0	[-10.8, -3.5]	0.0002	-7.0
	all	399	-9.8	[-13.8, -6.0]	0.0001	-10.0
node_count	components	400	+9.0	[+6.2, +12.0]	0.0001	+9.0
	clustering	400	+8.8	[+5.8, +11.8]	0.0001	+8.8
	rwse	400	+9.2	[+6.2, +12.2]	0.0001	+9.2
	degree	400	+8.0	[+5.0, +11.2]	0.0001	+8.0
	filler	400	+9.2	[+6.5, +12.2]	0.0001	+9.2
	all	400	+9.8	[+7.0, +12.8]	0.0001	+9.8
node_degree	components	400	+0.2	[+0.0, +0.8]	1.0000	+0.2
	clustering	400	-0.5	[-2.0, +0.8]	0.7246	-0.5
	rwse	400	-2.2	[-4.0, -0.5]	0.0240	-2.2
	degree	400	-6.5	[-9.2, -3.8]	0.0001	-6.5
	filler	400	-0.5	[-2.0, +0.8]	0.7232	-0.5
	all	400	-8.0	[-10.8, -5.2]	0.0001	-8.0

Table 10: qwen3-4b: all cells.

Task	Primer	n	Δ	95% CI	perm p	zeroed
connected_nodes	components	377	+0.5	[-0.5, +1.6]	0.6249	+0.5
	clustering	373	+0.0	[-1.3, +1.3]	1.0000	-0.8
	rwse	379	+0.0	[-1.3, +1.6]	1.0000	+0.5
	degree	376	+0.0	[-1.6, +1.6]	1.0000	+0.5
	filler	379	+0.8	[-0.3, +2.1]	0.3787	+1.2
	all	375	-0.5	[-2.1, +0.8]	0.7306	-1.2
cycle_check	components	381	+0.0	[+0.0, +0.0]	1.0000	-3.2
	clustering	394	+0.0	[+0.0, +0.0]	1.0000	+0.0
	rwse	395	+0.0	[+0.0, +0.0]	1.0000	+0.2
	degree	375	-0.8	[-1.9, +0.0]	0.2514	-5.5
	filler	396	+0.0	[+0.0, +0.0]	1.0000	+0.5
	all	392	+0.0	[+0.0, +0.0]	1.0000	-0.5
edge_count	components	60	+1.7	[+0.0, +5.0]	1.0000	+24.8
	clustering	59	+1.7	[-3.4, +6.8]	1.0000	+24.8
	rwse	49	+0.0	[-6.1, +6.1]	1.0000	+10.2
	degree	34	-23.5	[-38.2, -8.8]	0.0073	+8.5
	filler	50	-2.0	[-6.0, +0.0]	1.0000	+12.2
	all	41	-41.5	[-56.1, -26.8]	0.0002	+1.8
edge_existence	components	396	+0.0	[+0.0, +0.0]	1.0000	-0.5
	clustering	396	+0.0	[+0.0, +0.0]	1.0000	-0.8
	rwse	393	+0.0	[+0.0, +0.0]	1.0000	-1.2
	degree	398	+0.0	[+0.0, +0.0]	1.0000	+0.0
	filler	399	+0.0	[+0.0, +0.0]	1.0000	+0.0
	all	394	+0.0	[+0.0, +0.0]	1.0000	-1.0
node_count	components	400	+0.0	[-1.0, +1.0]	1.0000	+0.0
	clustering	399	+0.5	[+0.0, +1.3]	0.4936	+0.2
	rwse	399	+0.5	[+0.0, +1.3]	0.5005	+0.2
	degree	399	+0.5	[+0.0, +1.3]	0.5005	+0.2
	filler	400	+0.5	[+0.0, +1.2]	0.4870	+0.5
	all	400	+0.5	[+0.0, +1.2]	0.4870	+0.5
node_degree	components	380	-0.3	[-0.8, +0.0]	1.0000	+0.2
	clustering	386	-0.3	[-1.0, +0.5]	1.0000	+1.8
	rwse	381	-2.1	[-3.7, -0.8]	0.0072	-1.2
	degree	388	+0.0	[-0.8, +0.8]	1.0000	+2.5
	filler	386	+0.0	[+0.0, +0.0]	1.0000	+2.0
	all	383	-0.8	[-1.8, +0.0]	0.2466	+0.5

Table 11: qwen3-4b-think: all cells.